FLATNESS OF A SHEAFIFICATION FROM FLATNESS OF A COVER OF THE PRESHEAF OF MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one related lemma, recorded in assembly.external_references with its canonical tag as a navigation pointer.

Allowed standard prerequisites (editorial).

- Sheafification commutes with tensor products of presheaves of modules. Stacks Project, Modules on Sites, Lemma lemma-sheafification-tensor (tag 09BF). Identifies the tensor products of the sheafifications with sheafifications of presheaf tensor products, enabling the reduction.

Source reference. Original lemma 0GMW: sources/stacks-chow/tags.txt, source lines 20771–20771. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{O}$ be a presheaf of rings. Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$-modules. Assume that every object U of $\mathcal{C}$ has a covering $\{U_i \to U\}_{i\in I}$ such that $\mathcal{F}(U_i)$ is a flat $\mathcal{O}(U_i)$-module. Then $\mathcal{F}^\#$ is a flat $\mathcal{O}^\#$-module.*

Proof. Let $\mathcal{G} \subset \mathcal{G}'$ be an inclusion of $\mathcal{O}^\#$-modules. We have to show that

$$\mathcal{G} \otimes_{\mathcal{O}^\#} \mathcal{F}^\# \longrightarrow \mathcal{G}' \otimes_{\mathcal{O}^\#} \mathcal{F}^\#$$

is injective. By Lemma 0GMW the source of this arrow is the sheafification of the presheaf $\mathcal{G} \otimes_{p,\mathcal{O}} \mathcal{F}$ and similarly for the target. If U is an object of $\mathcal{C}$ such that $\mathcal{F}(U)$ is a flat $\mathcal{O}(U)$-module, then

$$(\mathcal{G} \otimes_{p,\mathcal{O}} \mathcal{F})(U) = \mathcal{G}(U) \otimes_{\mathcal{O}(U)} \mathcal{F}(U) \longrightarrow \mathcal{G}'(U) \otimes_{\mathcal{O}(U)} \mathcal{F}(U) = (\mathcal{G}' \otimes_{p,\mathcal{O}} \mathcal{F})(U)$$

is injective. Hence we reduce to showing: given a map of presheaves $f : \mathcal{H} \to \mathcal{H}'$ on $\mathcal{C}$ such that every U in $\mathcal{C}$ has a covering $\{U_i \to U\}_{i\in I}$ with $\mathcal{H}(U_i) \to \mathcal{H}'(U_i)$ injective, then $f^\#$ is injective. This we leave to the reader as an exercise. □

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